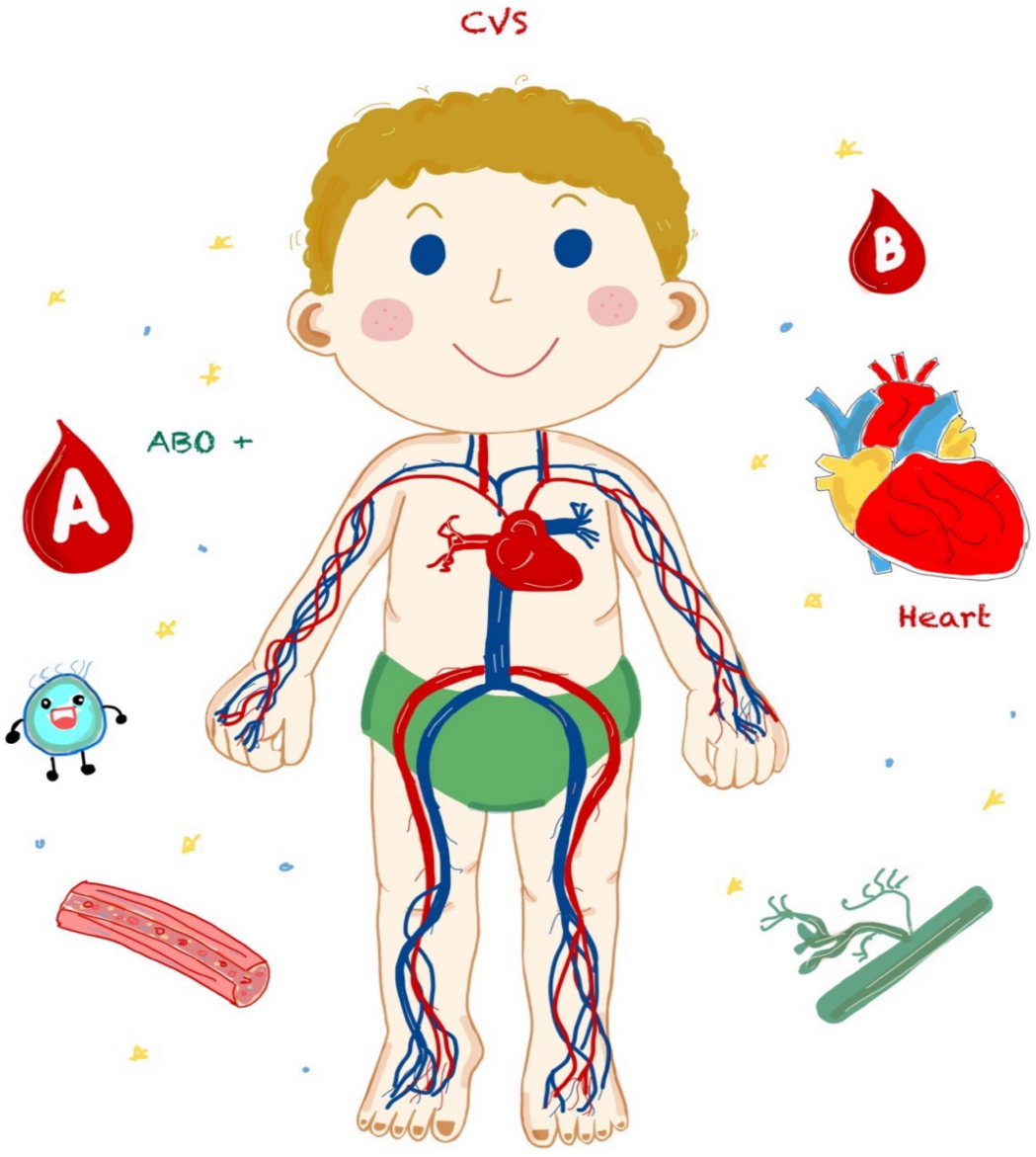
Paediatric cardiovascular system



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2024_2025

CONTENTS:

- Fetal circulation
- Murmur
- CHD etiology– classification
- A cyanotic CHD → VSD– ASD – PDA – AVC
  AS–COA– PS
- Cyanotic CHD → TOF– TGR –TAPVR
- Heart Failure
- Cardiomyopathy
- Rh Fever
- Infective endocarditis
- Arrhythmia– SVT
- Syncope

FETAL CIRCULATION

For the fetus, **Placenta** is the site of gas & nutrient exchange, so the lung do little work.

Shunts are necessary for survival:

1. **Ductus venosus** (bypass the liver)
2. **Foramen ovale** (RA → LA)
3. **Ductus arteriosus** (Pulmonary → Aorta)

Circulation:

- **Oxygenated** blood from placenta returns to fetus via 1 umbilical vein → Blood bypass liver via Ductus venosus to IVC → Mix with blood from lower limbs → This blood enters RA → **Pressure in RA > LA**

- Major part of this blood pass via the **Foramen ovale to LA** → Mix with deoxygenated blood from lungs at LA → goes to LV → Aorta → Coronary & Carotid arteries to supply heart & brain with relatively well oxygenated blood

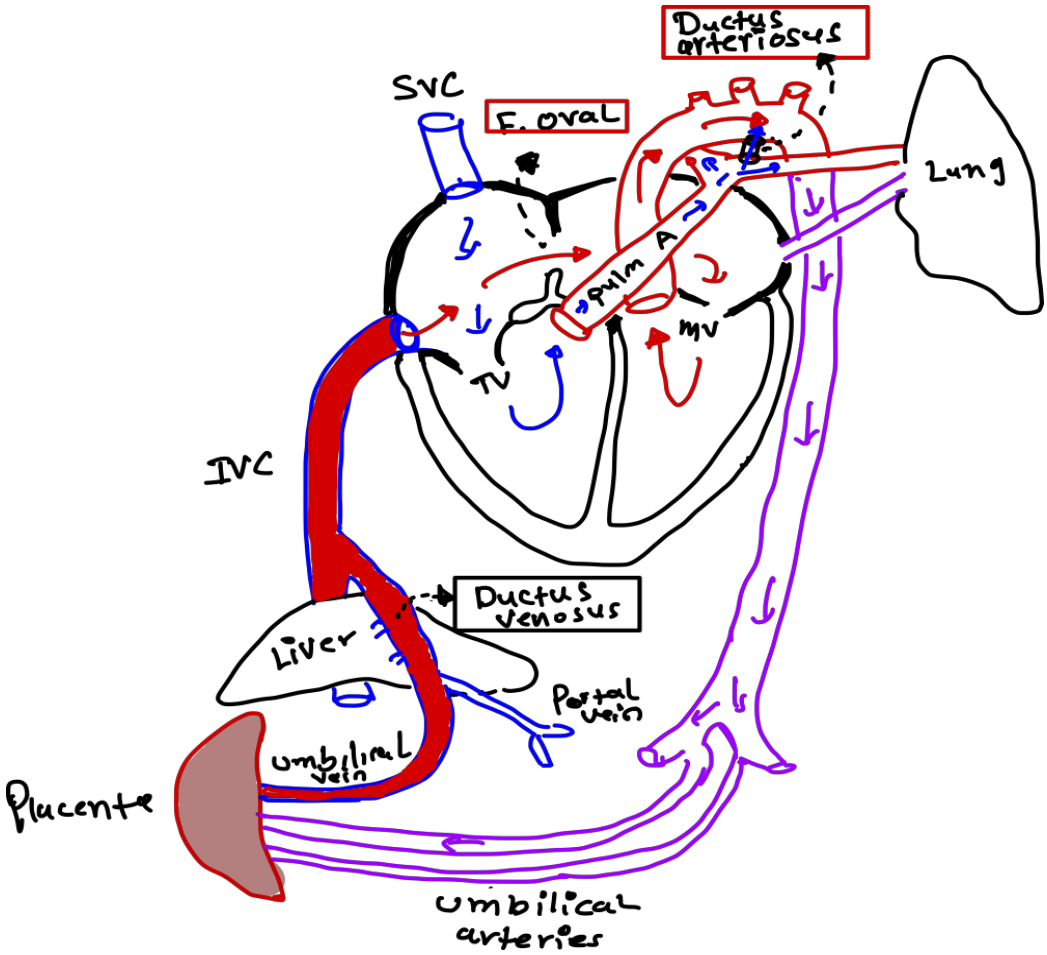
- **Deoxygenated** blood from SVC flows to RA → RV → pulmonary trunk
- Major part of its blood pass directly through **Ductus arteriosus** → descending aorta to supply lower half of body with relatively poor oxygenated blood

- Blood returns to the placenta via 2 umbilical arteries to repeat the circuit Changes occur at birth

With the first breath at birth:

1. Lungs expansion occurs → ↓ PVR → ↑ pulmonary blood flow → ↑ LA blood flow & pressure (LA pressure > RA pressure) → **closure of the foramen ovale**

blood flow in the LV → aorta → more oxygenated blood pass through the DA (from Lt to Rt) → leads to closure if DA (due to oxygenated blood → inhibited PG secretions by DA (also no more PG from placenta) → contraction of DA smooth ms → **functional closure of DA** (**anatomical** closure of DA Needs 2–3wks



Note:

- (PG KEEP → DA PATENT)
- (ANTI PG → DA CLOSURE)
- Antenatal (Rt side pressure is more than Lt side pressure)
- Closure of ductus arteriosus →تصبح (ligamentum arteriosum)
- Closure of foramen ovale → تصبح(fossa ovale)
- Closure of ductus venosus → تصبح(ligamentum venosum)

MURMUR

Murmur: Is turbulent of blood flow

- S_1 due to closure of M–T valves
- S_2 due to closure of A–P valves

Innocent murmur (Functional or Physiological)

Common [30% of children]

- No structural heart defect (Normal heart)
- Increase blood Flow through great vessels, "fever– anomia"

Criteria innocent: 10 S

1. Systolic (never diastolic)
2. Soft Grade (1 or 2/6 in intensity)
3. Short (early systolic)
4. Symptoms free (no cardiac symptoms)
5. Signs free (normal pulse– No thrill)
6. Site: Lt sternal edge, localized (No radiation)
7. Standing/Sitting (vary with position)
8. Special tests: Normal (ECG–X–ray– Echo)
9. Sounds (S_1 & S_2) Normal (musical or vibratory)
10. Still's murmur most common type

Treatment: REASSURANCE

Types of innocent Murmur:

Murmur	Age	site	origin
still's murmur {most common}	3–8 years	LLSB	Unknown (may be vibration of pulmonary valve
Venous hum	3–8 years	Infraclavicular	Jugular veins
Peripheral pulmonary stenosis (Pulmonary flow murmur)	newborn	Pulmonary area	pulmonary artery (valve)

Grades of cardiac murmur (Levine's Grading):

Intensity	Description
Grade I	Barely audible, very faint, only heard by an expert. No thrill (M <S1&S2)
Grade II	Audible, but soft. No thrill (M =S1&S2)
Grade III	Easily audible but No thrill (M > S1&S2)
Grade IV	Easily audible and associated with a thrill
Grade V	Easily audible, associated with a thrill, and still heard with the stethoscope only lightly on the chest(edge)
Grade VI	Easily audible, associated with a thrill, and still heard with the stethoscope off of the chest (1–2 cm)

CONGENITAL HEART DISEASE (CHD)

- Is the most common single group of structural malformation in infants
- Incidence: 8/1000 / {0,8%} Live born infants.
- About 30% require intervention in the first year of life, & about 1 in 10 stillborn infants have a cardiac anomaly

Causes of CHD: multifactorial.

Genetic	Environmental
1. Single gene defect 2. Familial 3. Chromosomal **The familial risk of CHD:** If siblings –> risk 4% If Father. –>risk 6% If mother –> risk 9% one of parents and sibling –> risk 20% if both parents have CHD –> risk Up to 30–50 %	1. Congenital infection: (TORCH)Congenital Rubella (PDA–PS 35%) 2. Maternal disease: A. Diabetes (HCM _VSD – TOF) B. SLE (complete heart block 35%) 3. Maternal Drugs as Alcohol (VSD, ASD) & Lithium (ebstein anomaly) Warfarin

Common syndrome and associated CHD:

Syndrome	ASSOCIATED CHD
• Trisomy 21 (Down syndrome) • Trisomy 18 (edward), 13 (patau)	• Atrioventricular septal defect(AVSD) 30%. • VSD & ASD (complex CHD) 80%
• Turner syndrome	• Bicuspid AV.coarctation of the aorta15%
• Noonan's syndrome	• Hypertrophic cardiomyopathyASD, pulmonary valve stenosis 50%
• Alagille syndrome • Williams's syndrome	• Pulmonary stenosis • Supravalvular aortic stenosis 85%
• DiGeorge syndrome	• Aortic arch anomalies, TOF, common arterial trunk 80%
• Fetal alcohol syndrome • Kartagner syndrome • Marfan syndrome	• ASD, VSD, TOF 25% • Dextrocardia • Aortic artery dissection, MVP

Congenital heart disease classification

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Acyanotic CHD	Cyanotic CHD
1. with left to right shunt:	1. TOF (tetralogy of fallot)
A. VSD (Ventricular septal defect) 30% mc	2. TGA (transposition of great arteries)
B. ASD (Atrial septal defect)	3. TA (Tricuspid atresia)
C. PDA (Persistent ductus arteriosus)	4. TAPVR (Total anomalies of pulmonary venous return)
D. AVSD (Atrio–ventricular septal defect) partial type	5. TA (Truncus arteriosus)
2. with obstruction:	6. Pulmonary atresia PA
A. Aortic stenosis AS	7. Single ventricle
B. pulmonary stenosis PS	8. Hypoplastic lt heart syndrome
C. Coarctation of aorta CoA	9. Ebstein m anomalies

Note:

- The Most Common CHD is VSD 30%
- The Most Common Cyanotic CHD in children is TOF
- The Most Common Cyanotic CHD in Neonate is TGA

Clinical Presentation of CHD:

- Asymptomatic
- murmur
- Heart Failure
- Cardiogenic Shock
- Cyanosis.
- FTT /Recurrent chest infection
- Antenatal diagnosis by U/S
- Arrhythmia.

VENTRICULAR SEPTAL DEFECTS(VSD)

- Defect in the interventricular septum
- It's the most common CHD ≃30%

Classification

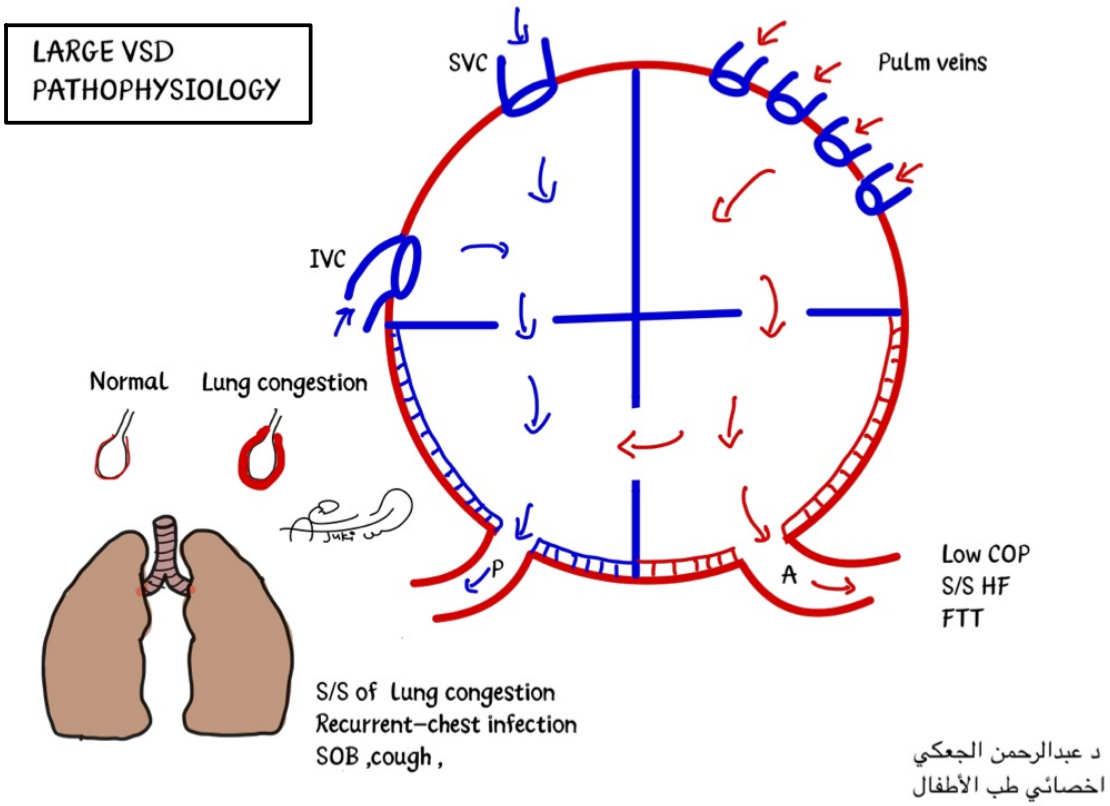
Site	Size
1. Peri membranous USD ≃70% 2. Muscular VSD ≃5.20% • Single • Multiple "Swiss Cheese"	1. Small VSD ≤ 3mm (commonest). <AV ≃80% 2. Large VSD ≥ AV ≃10–20%

Clinical Picture:

It depends on: 1. Size. 2. Pressure gradient between ventricles.

SMALL VSD	LARGE VSD
• Asymptomatic • Delayed diagnosis usually **Clinical picture:** • Discovered accidentally with PSM murmur	• Symptomatic • Early diagnosis usually **Clinical picture:** 1. S/S of decreased COP (heart failure) breathlessness, sweating, FTT usually started >1wk old . ? Why? 2. S/S of increased pulm blood flow

Examination: • Normal wt (No FTT) • No signs of HF • Loud pansystolic murmur at lower left sternal edge (smaller defect →louder murmur)	→ (lung congestion) recurrent cough & chest infection **Examination:** • Signs of HF (Tachypnoea, Tachycardia and Tender enlarged liver) • Under wt • Soft pansystolic murmur or no murmur (due large defect) • But may hear ejection systolic murmur due functional pulmonary stenosis or Apical mid-diastolic murmur (from increased flow across the pulmonary & mitral valve) • Loud S2 (bad sign –plum hypertension develops)
Investigation: Chest x ray —> Normal. ECG ———>Normal Echo ———>diagnostic (detect anatomy, No pulmonary hypertension)	**Investigation:** Chest x ray —> Cardiomegaly Plethoric lung, Pulmonary oedema ECG ——> biventricular hypertrophy early Echo ———> diagnostic (detect anatomy, ± high pulmonary hypertension)
Treatment: • Small VSD will close spontaneously in 1st 2–3 years (especially muscular type) • Follow up (every 3–6 months) clinically &echocardiography assessment • Prophylaxis against IE	**Treatment:** • treatment of HF (decongestive therapy diuretic + captopril) • Early surgical correction (3–6 months) open or Cath to prevent pulm hypertension, heart failure and faltering growth • Prophylaxis against IE
Complications: 1. Infective endocarditis	**Complications:** 1. Heart failure 2. FTT 3. Recurrent lower RTI 4. IE 5. Eisenmenger syndrome[late] 6. Arrhythmia


LARGE VSD
PATHOPHYSIOLOGY
SVC
Pulm veins
IVC
Normal
Lung congestion
P
A
Low COP
S/S HF
FTT
S/S of Lung congestion
Recurrent-chest infection
SOB ,cough ,
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Eisenmenger syndrome

Def: Pulmonary vascular diseases complication of large Lt–to–Rt shunt (e.g: VSD, ASD, PDA)

It's irreversible reversal of shunt (Become Rt to Lt shunt) due to increased pulmonary vascular resistance.

Clinical picture

- Onset: Usually in the 2nd or 3rd decade of life.
- Cyanosis becomes apparent & progressive
- dyspnoea, chest pain, haemoptysis, clubbing
- risk of arrhythmias

Cardiac examination:

- RVH
- Visible pulsations & palpable S2 over the pulmonary area
- Loud S2 by auscultation.

Investigations:

- CBC: Polycythaemia due to hypoxia
- CXR: Cardiomegaly & oligemic lung field.
- ECG: RVH& tall spiked P wave (P pulmonale)
- Echo: Underlying CHD–RVH
- Doppler studies: Detect direction of shunt & estimate the pulmonary pressure
- Cardiac catheterization: for: Pressure measurement, also measure the Response to VD therapy (NO, prostacyclin)

Treatment

- Medical: pulm VD as (Sildenafil+Viagra)(phosphodiesterase inhibitor)(Inhaled Nitric Oxide (NO)
- Surgical: (heart lung transplantation)

Prevention

Early surgical ttt of large intracardiac defects during infancy.

ATRIAL SEPTAL DEFECT (ASD)

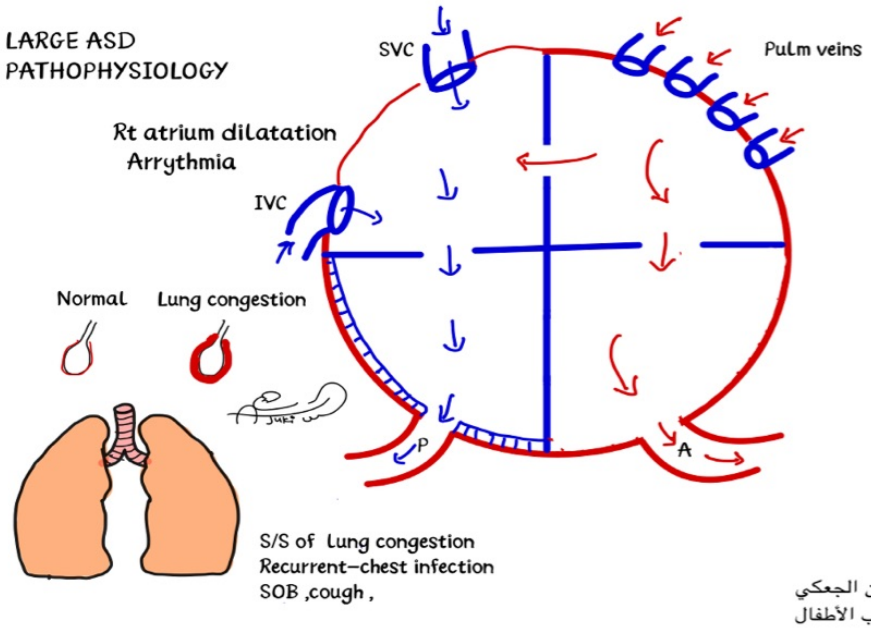
- Defect in the interatrial septum.
- % = 7% More in ♀.

Types of ASD:

Ostium secundum = ASD secundum	ostium primum= ASD primum	Sinus Venosus
Most common 75–80 %	2nd common 15–20%	Least common
in the middle part near foramen oval	in the lower part near mitral valve	in the upper part near SVC
Less serious	more serious (associated with AV Valve or mitral regurgitation) → partial atrioventricular septal defects (AVSD)	may be associated with PAPVR

Clinical Picture: (Lt to Rt shunt with increase PBF)

- Asymptomatic "commonly "
- Recurrence Chest infection due to Lung congestion.
- Arrhythmia (4th decade onwards).



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Examination:

- No murmur due to ASD. **MCQ**
- Ejection systolic murmur in the ULSB Due to increased B1 Flow across pulmonary valve & Mid diastolic murmur relative tricuspid stenosis
- **Fixed Wide splitting S_2** (often difficult to hear) due to the right ventricular stroke volume being equal in both inspiration and expiration.

Note: With a partial AVSD, an apical pansystolic murmur from atrioventricular valve regurgitation.

Investigation:

1. Chest X Ray → Cardiomegaly – Plethoric Lung
2. ECG → ASD secundum – Normal or RVH, Right axis deviation ± RBBB.

   → ASD Primum – RVH – Superior or left axis deviation, LBBB
3. Echo → Diagnostic

Treatment:

- ASD large cause Rt dilatation needs treatment (Small ASD → Spontaneous closure)
- ASD secundum – catheterization closure with occlusion device.
- ASD primum – surgical correction at 3–5 years

Complication:

1. HF (**rare)**
2. FTT
3. IE (**No risk of IE in ASD secundum**)
4. Recurrent Chest infection
5. Arrhythmia
6. Eisenmenger syndrome

PATENT DUCTUS ARTERIOSUS (PDA), Persistent ductus arteriosus

Called **Persistent ductus arteriosus**, If Ductus arteriosus **fails** to close by **1 month** after the expected date of delivery.

PDA is more in:

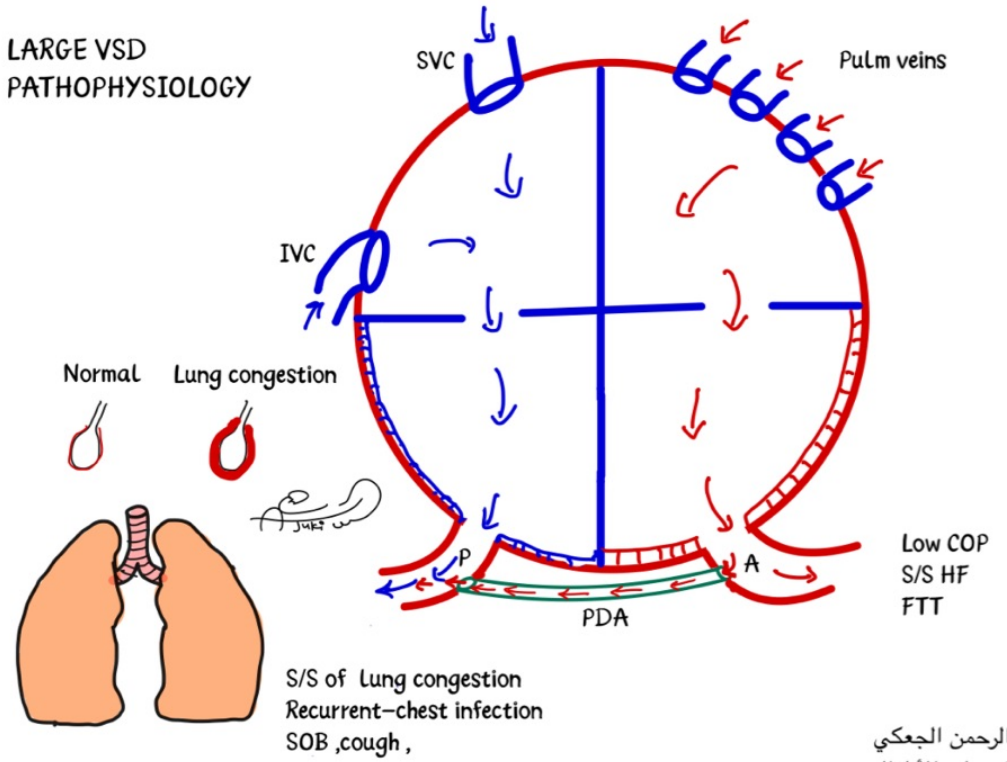
- Preterm baby (PDA is not CHD in preterm).
- Congenital Rubella Syndrome
- Birth asphyxia

Epidemiology:

- % ≃ 12%
- Sex: more in ♀

Clinical Picture: (Lt to Rt Shunt with increased PBF)

1. Small PDA "< 4mm "
   - Asymptomatic
   - No FTT or HF.
   - Murmur (continuous murmur in Lt infra-clavicular area)
   - Echo → diagnostic
2. Large PDA.
   - Symptomatic
   - S/S HF – FTT
   - S/S Pulmonary congestion (recurrent chest infection)



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ملاحظة :هذه السلايد توضيحية فقط .

Examination (Large PDA):

- General → poor Weight gain (FTT)
- Pulse
  - Large volume pulse
  - Bounding pulse (collapsing)
  - Wide pulse pressure (More escape of blood during diastole → ↓ SBP, ↓↓↓↓ DBP → hyper dynamic circulation)
- Precordial examination
  - Machinery (continuous) Murmur in Lt infra clavicular area ± thrill
  - Other (Apex → Relative MS: Mid diastolic murmur, Aortic area → Relative AS: Ejection systolic murmur)

Investigation:

- Chest X Ray → Cardiomegaly – plethoric lung if large PDA.
- ECG → LVH then Rt
- Echo →Diagnostic

Treatment:

1. **In Preterm baby**: medical ttt in 1st wk → **Anti PG (Indomethacin or Ibuprofen) + restriction of IVF.**

   S/E of drug: NEC, Bleeding, Renal failure

2. **In full term infant**:

   Small PDA → Trans-catheter closure: using coils or duct occluded (method of choice) in 1st year

   Large PDA → Surgical ligation: (risk of IE)

Complication of Large PDA: As Large VSD

- Heart failure
- FTT
- Recurrent lower RTI
- IE (even in small PDA there's risk of IE)
- Eisenmenger syndrome[late]
- Arrhythmia

	VSD	ASD	PDA
%	≈30%*	≈7%	≈12%
Sex	Equal	More in ♀	More in ♀
Hemodynamic	Lt to Rt shunt with increase PBF	Lt to Rt shunt with increase PBF	Lt to Rt shunt with increase PBF
Types	**Site:** • Perimembranous • Muscular **Size:** • Small VSD • Large VSD	**Site:** • Ostium Secundum • Ostium primum • Sinus venosus	**Size:** • Small • Large
Clinical picture	Small Size: Asymptomatic Large size: • Symptomatic • S/S HF • S/S Lung congestion	• Asymptomatic usually • S/S lung congestion • Arrhythmia	Small: Asymptomatic Large: • Symptomatic • S/S HF • S/S Lung congestion
Hypertrophy	Biventricular hypertrophy	Right ventricular hypertrophy –RAD	Left ventricular hypertrophy–LAD
Physical findings clinically	**Small** VSD PSM in LLSM radiated all over the precordium **Large** VSD: S/S HF –FTT PSM in LLSM	**No** murmur due to ASD defect **Fixed wide splitting S2** * ESM at ULSE	**continues** murmur Lt infraclavicular area **Pulse** **Large** volume pulse **Bounding** pulse (water hammer) **Wide** pulse pressure
Investigation	• Chest X Ray: cardiomegaly, plethoric lung • ECG → BVH • Echo → diagnostic	• Chest X Ray: cardiomegaly, plethoric lung • ECG → RVH, RBBB • Echo →diagnostic	• Chest X Ray cardiomegaly, plethoric lung • ECG→LVH, LBBB • Echo →diagnostic

TREATMENT	**Small**: Spontaneous closure **Large**: Diuretics, captopril, calories & Surgical repair at 3–6 m	**Small**: spontaneous closure **Large secundum**: cath closure 3–5yr **Large premium**: surgical closure at 3yr.	**Preterm**: Anti PG –restrict iVF Term **infant**: Cath closure or surgical ligation
Complication	If large VSD 1. HF 2. IE (in small also) 3. FTT 4. R Chest infection 5. Eisenmenger syndrome 6. Arrhythmia	IF large ASD: 1. HF (rare) 2. IE (Not in secundum) 3. FTT 4. R Chest infection 5. Eisenmenger 6. Arrhythmia	If large PDA: 1. HF 2. IE (in small also) 3. FTT 4. R Chest infection 5. R Chest infection 6. R Chest infection

ATRIOVENTRICULAR SEPTAL DEFECT (AVSD)
(Endocardial cushion defect ECD)

- **AVSD** Associated commonly with ***Down syndrome (Trisomy 21)***

TYPES:

Partial AVSD (Primum ASD)	Complete AVSD (blue and breathless)
1. Ostium Primum defect 2. Abnormal AV valve (mitral regurgitation) **Clinical picture:** As ASD primum **Treatment** As ASD	1. ostium primum 2. VSD inlet 3. common AV valve (5 leaflets single valve) **Clinical picture:** • cyanosis at birth or heart failure at 2 –3 weeks of life–FTT • No murmur heard • It May Detected in routine echocardiography screening in a newborn baby with Down syndrome **Treatment** • treatment of HF • surgery at 3–6 months of age

COARCTATION OF AORTA (COA)

- Localized **narrowing** of the aorta
- **Site:** any point from the arch down to iliac bifurcation.
- **Common site:** just distal to the origin of the **Lt SCA**. (78% Juxta ductal)
- **More Common in ♂** (male)
- Common CHD in **Turner syndrome.**
- It may be associated with other congestion anomalies as bicuspid aortic valve
- **Hemodynamic:**
  - increase BP in the Upper Limbs "upper Parts of the body"
  - decrease BP in the Lower Limbs "lower Parts of the body"
  - Lt–V–Hypertrophy.
  - Collaterals develop between proximal & distal aorta.
  - In *sever COA+ PDA (newborn):* perfusion of Lt limbs is dependent on the duct → leads differential cyanosis (nipple up pink, nipple below bluish)

C/P: it depends on severity of coarctation

Sever COA in the sick infant:

C/P

- Circulatory collapse at 2nd days of age when the duct closes.
- Heart failure.
- Poor Lower limbs perfusion, weak or absent femoral pulse.
- Severe metabolic acidosis.

Investigation

- Chest X Ray: Cardiomegaly 'HF'
- ECG: Normal
- Echo: diagnosis
- CT angiography

Treatment: In severe COA Sick infant:

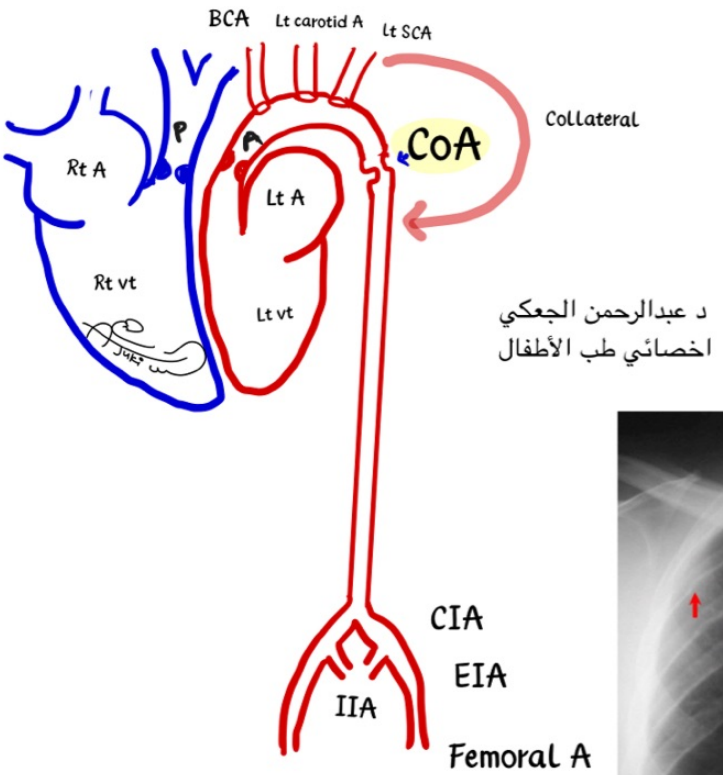
1. Resuscitate (ABC).
2. Prostaglandin 'Early'.
3. Referral to cardiac centre for early surgical intervention.

Milder COA (Adult type COA)

C/P

- Asymptomatic.

- S/S hypertension (older age) Headache – Epistaxis – Dizziness.
- Radio Femoral delay (Due to collaterals)
- Ejection systolic murmur at upper sternal edge.
- Collaterals heard with continuous murmur at the back between scapula.



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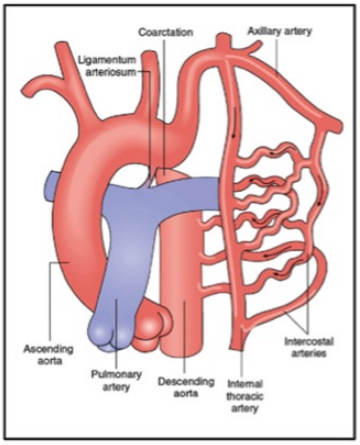
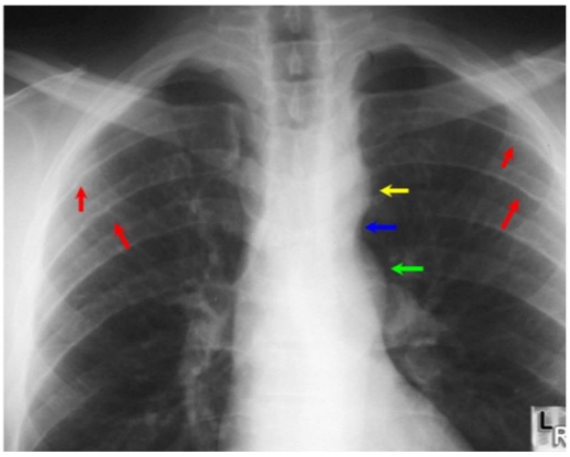




Investigation

- Chest X Ray:
  1. **Rib notching ">6y"**, "Due to collaterals (Rosler sign)
  2. **3 signs** with visible notch in the descending aorta at site of coarctation
- ECG → LVH
- Echo→ diagnosis, any associated CHD (Echo Doppler: The best)
- Catheterization: Preoperative to detect site & severity
- CT or MRI angiography.

Treatment: adult type CoA

- **Mild CoA**: Medical Control of HTN, heart failure, & IE & regular flow up.
- **Moderate CoA**: Trans-Catheter Balloon dilatation ± Stent.

- **Severe CoA**: Surgical repair, Co-artectomy (in early childhood to avoid persistent HTN)

Complication:

1. Systemic HTN complications.
2. IE (prophylaxis)
3. Lt Vt failure.
4. Cerebral aneurysm → rupture → Intracranial haemorrhage.
5. Sudden death.

Important Note

- Normal person {BP in the lower limbs > BP in the upper limbs}
- In **CoA** patient {BP in the **upper** limbs > BP in the **lower** limbs}
- (That why if you suspect a case with CoA: you should measure BP in the 4 limbs)

	Aortic Stenosis (AS)	Pulmonary Stenosis (PS)
%	≃5%	≃7%
syndrome associated	**Williams** Syndrome	**Noonan's** Syndrome, CRS.
Sex	More in **boys**	More in female
Types	1. Valvular** (RH fever-congenital) 2. Supravalvular (**Williams** Syndrome) 3-Subvalvular (congenital)	1. Valvular** 2. Subvalvular: in TOF 3.Supra-valvular: Fibrous ring above valve
Hemodynamic	AS → Obstruction of left ventricular outflow → Low COP. Pressure overload → LVH → Left ventricular failure.	PS → Obstruction of right ventricular outflow →RVH →Rt side HF .
Clinical Picture	Mild + Mod AS 1. Asymptomatic 2. ESM **Mod to Severe AS** *Easy Fatigability	Mild to mod PS 1. Asymptomatic 2. ESM Moderate: breathlessness

	*Exertional chest pain *Syncope *Sudden death Newborn & critical AS: HF – Shock	Severe, Critical: Cyanosis + HF early in life (in newborn)
Examination	General Weak slow rising pulse– decrease BP Pericardium: *Heavy apex beat *Carotid thrill* *ESM in 2nd ICS Radiating to neck (carotid) *Apical ejection clicks (valvular type).	Pericardium: *Lt Parasternal heave (RVH) *ESM in 2nd ICS (Lt) radiated to back *Ejection clicks best heard at the upper left sternal edge.
Investigation	**CX Ray** →Cardiomegaly **ECG** → Lt–V–H **Echo** →Diagnostic	**CX Ray** → cardiomegaly **ECG** → Rt–V–H **Echo** → Diagnostic
Treatment	**Asymptomatic:** (Flow up+ echo +IE prophylaxis) **Symptomatic**: 1. balloon valvotomy children 2. Aortic valve replacement in newborn with severe AS	**Asymptomatic:** (Flow up+ echo + IE prophylaxis) **Symptomatic:** Transcatheter balloon dilatation is the treatment of choice in most children
Complication	1. HF 2. IE 3. Sudden Death	1. HF 2. IE

CYANOTIC CHILD

Cyanosis: 2 types

- Peripheral cyanosis: (**sites:** hands and feet, or around the mouth, tip of nose, ears lobules), caused by hypothermia, shock, low COP, associated with **normal** arterial saturation
- Central cyanosis: (**sites:** tongue, soft palate) appears Clinically if reduced Hb **exceeds**(*50g/L=5g/dL*), caused usually by respiratory or cardiac disease, associated with **low** arterial saturation. (it is Less pronounced in anaemic child)

Management of the cyanosis newborn

If you have a **newborn** with **central** cyanosis and have **low** oxygen saturations ≤94%, to determine the **cause** of cyanosis: is it **cardiac** or **respiratory** disease?
You should do:

Central cyanosis

Hyperoxia Test (Nitrogen washout test):

The infant is placed in 100% oxygen (headbox or ventilator) for 10 minutes.: [Look for O2 Saturation or partial pressure of oxygen (PaO2) changes in ABG].
Result:

1. If the O2 Saturation or right radial arterial partial pressure of oxygen (PaO2) remains low (<113mmHg) after this time → diagnosis of 'cyanotic' congenital heart disease

2. If the O2 Saturation or right radial arterial partial pressure of oxygen improved → respiratory disease or persistent pulmonary hypertension

Note (ABG (PaO2) must be performed as oxygen saturations are not reliable enough in this range).

Immediate management:

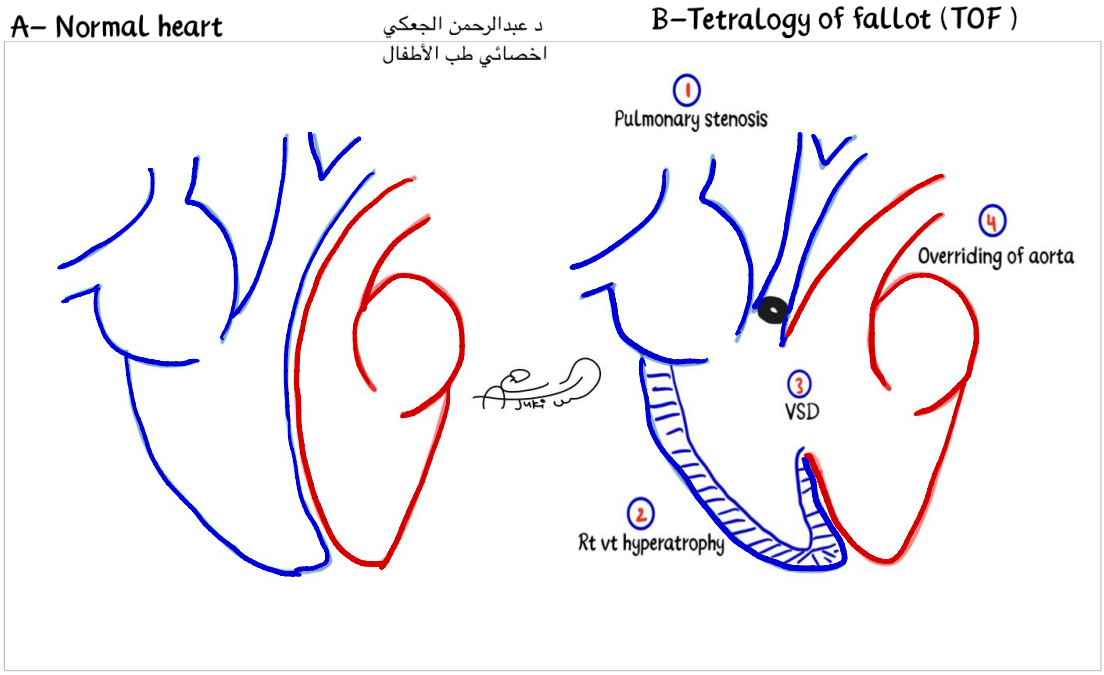
1. Stabilize the airway, breathing, and circulation (ABC), with mechanical ventilation if necessary.
2. Start prostaglandin E infusion (5 ng/kg per min).
   side-effects of PG (apnoea, jitteriness and seizures, flushing, vasodilation and hypotension)
3. urgent cardiac consultation, echocardiography & surgical repair

TETRALOGY OF FALLOT (TOF)

- Is the **most common** cyanotic CHD in children % ≃5% –10%

Components of TOF: (4)

1. **Pulmonary stenosis** (PS): Infundibular (commonest) or sub valvular
2. **VSD**: big size
3. **Right ventricular hypertrophy (RVH)**: mild (↑ with time → ↑ PS)
4. **Overriding Aorta**: receive blood from both ventricles



Haemodynamic:

*During RV contraction, most of blood will be pushed to LV through VSD → cyanosis

*Few amounts of blood is pushed through stenotic pulmonary valve → lung oligemia
***According to degree of pulmonary stenosis

Severe PS → early right to left shunt → early (Neonatal cyanosis), rare.

Mild to Moderate PS → ↑ stenosis with time → cyanosis after 6 months or later (common)

Clinical picture: it depends on the degree of PS.

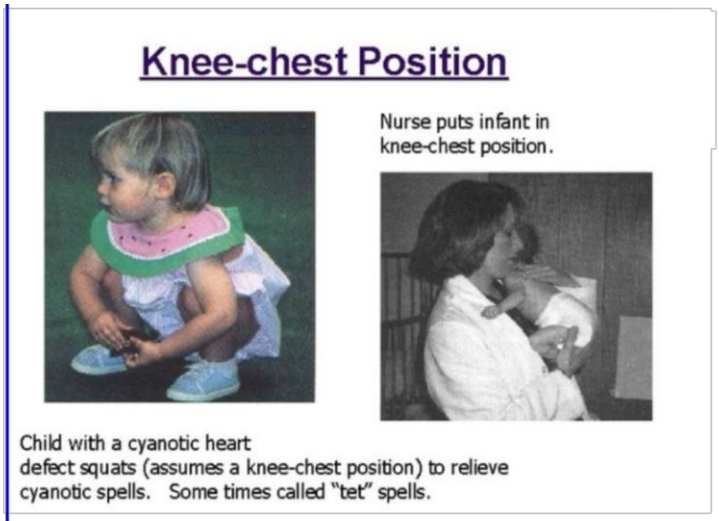
1. Cyanosis: Usually not at birth, as hypertrophy of the pulmonary infundibulum occurs as the patient grows (3–6 months)

•May be early (neonatal cyanosis in the first few days of life) if marked RV outflow obstruction (the pt. is duct–dependent)

2. Clubbing: in long–standing cases due to hypoxia

3. FTT: failure to gain wt due to hypoxia

CHILD WITH TOF
1–CYANOSIS
2–CLUBBING OF FINGERS



Squatting position in tet spells

4. Tet spells or cyanotic /hyper cyanotic or hypoxia spells (EMERGENCY)

Definition: attack of cyanosis & RD in infants and children with cyanotic CHD(TOF) caused by spasm of the pulmonary infundibulum → more decrease in the pulmonary blood flow.

Its **emergency** may lead to myocardial infarction, cerebrovascular accidents and even death if left untreated

Clinical picture (occurs at any time but usually in the morning during crying or playing)

- Cyanosis or increased cyanosis.
- Rapid breathing, RD
- Agitation– irritability, inconsolable crying (hypoxia)
- Squatting position (knee chest position)
- Pallor – Severe hypoxia, acidosis
- Convulsions, coma, death

Treatment of Tet spells: مهمة جدا

Usually, self–Limiting followed by a period of sleep, If prolonged >15m needs R_x

1. Squatting position (to ↓SVR & thus ↑ pulmonary blood flow)
2. O2!
3. morphine 0.1 mg/kg SC, IM (↓ irritability)
4. IV propranolol 0.1 mg/kg slow IV (↓infundibular spasm)
5. α adrenoceptor agonist "Phenylephrine" (VC)
6. NaHCO3 1 ml/kg slow IV (correct acidosis)
7. IV volume administration
8. ETT+MV.

- **The murmur during Tet spells → (either disappear or decrease in intensity)**
- **During management Usually keep the infant calm & Delay unnecessary blood sampling.**
- **Diagnosis of TOF can be also (antenatally or by identification of a murmur in the first 2 months of life)**

Examination of TOF:

General: Central Cyanosis, FTT, Clubbing of the fingers and toes in older children.

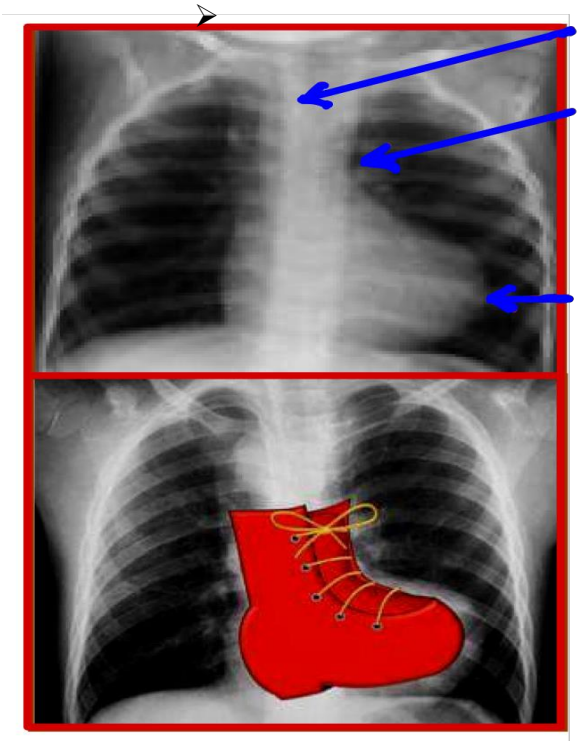
Precordial: loud ejection systolic murmur at the left sternal edge.

Single S2(Aortic component only, absent Pulmonary component)

Lt parasternal heave (RVH)

Investigation:

- **Chest X Ray**
  - Shape boot shaped (Coeur en sabot) → up lifted apex & exaggerated waist
    Size Relatively small
  - oligemic lung



- ECG: RVH (older child)
- **Echocardiography** → diagnostic

- **cardiac catheterization** (or MRI scan) may be required to show the detailed anatomy of the coronary arteries if not seen on echocardiography.
- CBC: polycythaemia

Treatment of TOF:

Medical Rx	Surgical Rx
• Rx of Tet spells. • Prophylaxis IE.	A. **Palliative** (***Blalock Taussig "BT" shunt***–>SCA +ipsilateral PA) done early. Idea: artificial shunt between Ao & PA → ↑ PBF → ↑ oxygenation → improve growth Other types of shunts: **Waterston** (ascending aorta + Rt PA) & **Potts** (descending aorta + Lt PA) B. **Definitive Total correction** at age (6–12m) –>Closure of VSD & repair of RV outflow obstruction.

Complication of TOF. MCQ

1. Infective endocarditis "IE"
2. Cyanotic Spells.
3. HF (rare).
4. FTT (short stature)
5. Cyanosis: decreased O_2 → ↑ Erythropoietin → increase RBC (**Polycythaemia**)
6. Thromboembolic complications
7. Brain abscess: due to lack of filtration by pulmonary circulation → paradoxical embolism (hypoxia → anaerobic organisms).
8. Brain infarction due to Hyper cyanotic spells → convulsions
9. Brain thrombosis: due to polycythaemia

TRANSPOSITION OF GREAT ARTERIES 'TGA'

- **Aorta** arises from RV & **Pulmonary** artery arises from LV
- 2 Parallel circulations (The blue blood is returned to the body and the pink blood is returned to the lungs)
- Communication is a **must** (B\C Not compatible with life)
- Mixing of blood by through a shunt (PFO, VSD or PDA)
- PDA is essential → if closed death may occur (duct dependent)

Clinical Picture:

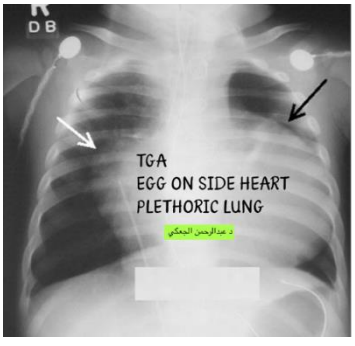
1. Cyanosis – usually on day 2 after birth (why ???).
2. S/S HF.

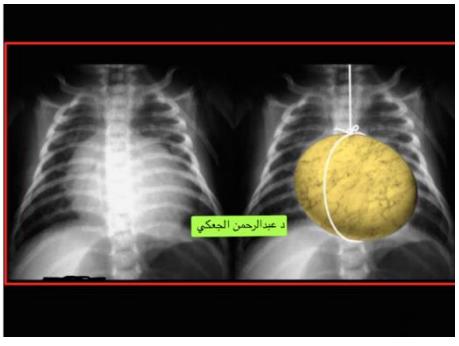
Examination:

- Cyanosis (always)
- No murmur due to transposition
- S_2 Loud & Single

Investigation:

- Chest X Ray
    1. Shape: ***Egg on side***
    2. Size: ***cardiomegaly***
    3. ***Plethoric*** lung (due to increased PBF)
- ECG Normal Usually
- Echo: diagnostic



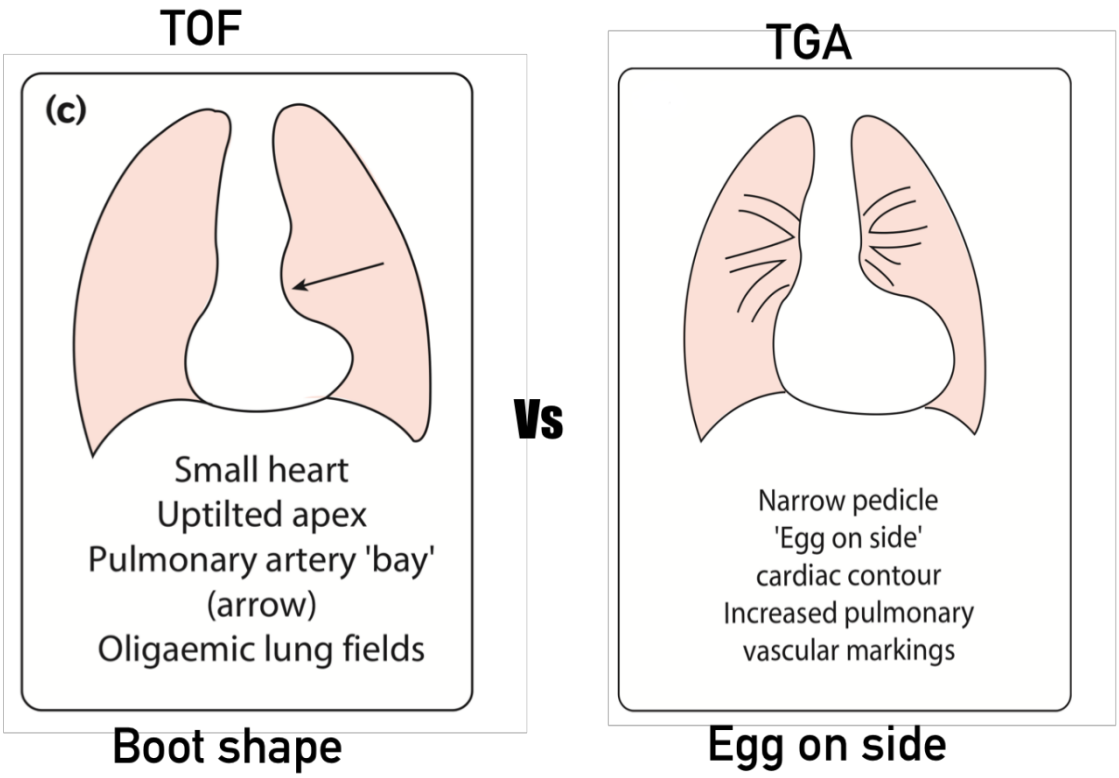


Treatment:

1. ***Sick Cyanotic neonate***
   - PG infusion
   - Trans catheter Balloon atrial septostomy (***Rash kinds*** operation)
2. ***Total definitive***
   - Surgical correction (arterial switch operation) (early should be done)

Types of TGA:
1- Isolated TGA 2- TGA + VSD
3- TGA + VSD + PS 4- L-TGA = TGA +ventricular inversion

Chest X ray findings of TOF & TGA:

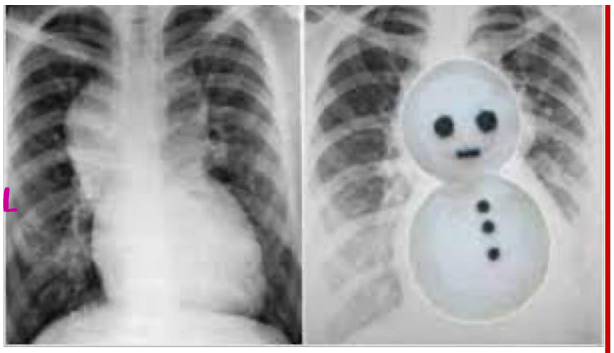


COMPLEX CHD

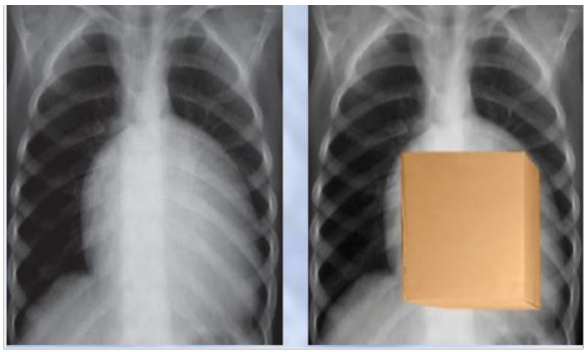
is difficult to generalize about these conditions (tricuspid atresia, mitral atresia, double inlet left ventricle = single ventricle, common arterial trunk = truncus arteriosus) because their main presenting feature depends on whether cyanosis or heart failure is more predominant. ***Tricuspid atresia*** is the most common.

CHD	Defect	Presentation	Treatment
Tricuspid Atresia	• Tricuspid atresia (ASD) • Blood pass from RA → LA → LV "enlarged"	• Early-onset cyanosis • ECG: LT axis deviation in ECG (only the left ventricle is effective, the right being small and non-functional)	Medical: PGEI +supportive. Raskind operation Surgical: Palliative (BT shunt) pulm artery banding or Glenn or Fontan
Pulmonary atresia either with or without VSD.	• Pulmonary atresia • Blood pass from RV →LV →aorta • Duct dependent	• Early-onset cyanosis • No murmur • ± HF	Medical: PGEI + supportive Surgical: Palliative (BT shunt) or total correction
Truncus arteriosus	Single trunk → aorta & PA	• Cyanosis → variable onset • Features of ↑ pulm bl flow	Urgent surgical correction in infancy
Ebstein anomaly	• Downward displacement of the tricuspid valve • Huge RA + Small RV • Rt to Lt shunt (PFO)	• Variable-onset cyanosis • Murmur (TR) • Arrhythmias (SVT)	Medical: PGEI + supportive + anti-arrhythmic drugs Surgical: Rarely needed
Total anomalous pulmonary VR (TAPVR)	• All pulmonary veins are not connected to LA Types: 1. Supra cardiac (50%): into SVC 2. Cardiac (25%): into coronary sinus or RA 3. Infra cardiac	• Variable-onset cyanosis Early usually • ± HF signs	Medical: PGEI + supportive Surgical: Rashkind or total correction

	(20%): into PV, ductus venosus, HV or IVC 4.Mixed type (5%) • associated with ASD • Duct dependent		
Single ventricle {double inlet left ventricle}	• Absent interventricular septum → common ventricle	• Variable–onset cyanosis Early usually	supportive mainly
Hypoplastic left heart syndrome (HLHS)	• Variable degrees of hypoplastic LA, LV & ascending aorta • Mitral & aortic stenosis • Duct dependent	• Early–onset (may be differential) cyanosis • decreased COP manifestations & absent pulsations • Usually No murmur • very sick, acidosis collapse & Death in the 1st month	Medical: PGEI + supportive Surgical: Norwood Cardiac transplantation



TAPVR —> Snowman appearance



Ebstein anomaly —> Box shaped heart

HEART FAILURE 'HF'

Definition: inability of the heart to maintain adequate cardiac output to meet the metabolic needs of the body

Causes (Congenital or Acquired):

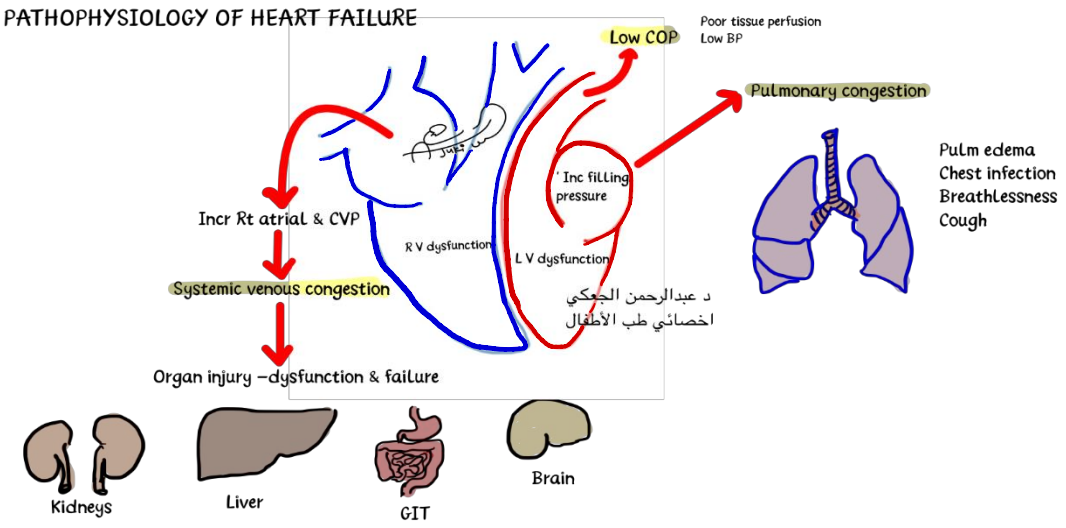
1. Obstructed Systemic circulation (Neonate)
2. Lt to Rt shunt with high PBF (Infants)
3. Acquired disease (older children adolescents)

According to the Age:

Fetal	Preterm	Term	Infants and toddlers	children & adolescents
• Anaemia: Rh incompatibility • Arrhythmias: SVT, VT, Complete HB	• Large PDA, VSD • fluids overload (IVF)	• Severe Lt vt obstruction as Severe CoA ,AS ,HLHS, Interrupted Aortic Arch. • Acute viral or ischemic myocarditis • SVT, others TGA, TA.	• Large Lt to Rt shunt as VSD ,PDA ,AVSD. • myocarditis, cardiomyopathy. • SVT, KD	• RHD, KD. • IE • myocarditis, cardiomyopathy • HTN, Eisenmenger Syndrome

Others: **High COP failure** (hyperdynamic circulation): Anaemia /AV fistula/ Thyrotoxicosis.

Pathophysiology:



C/P:

1. **Symptoms**
   - Breathlessness (increase with feeding or exertion)
   - Sweating – Due to increase CA release
   - Poor feeding – interrupted Feeding
   - Recurrent Chest infection
   - FTT

2. **Signs: (3T– 1C) or (2 Tachy, 2 Megaly)**
   1. Tachycardia *2. Tachypnea*
   3. Tender Enlarged Liver. *4. Cardiomegaly on C X Ray*

   Other important signs:
   - FTT
   - Gallop Rhythm (S_3 + Tachycardia)
   - Sweating & Cool peripheries
   - Bilateral basal crepitation due to pulm oedema.

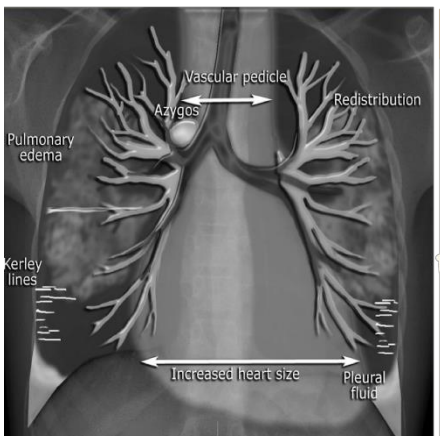
> **Note:** مهم
>
> (Ankle oedema – orthopnoea – PND, Congested neck veins JVP → Are not signs of HF in infancy) it's seen commonly in adolescents and adult adults

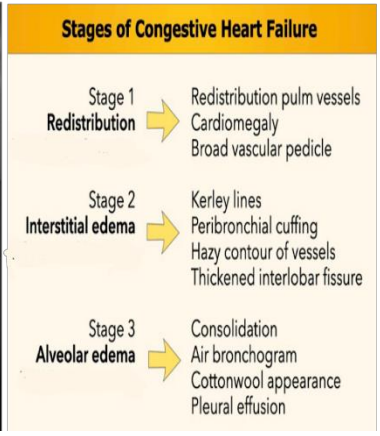
Diagnosis:

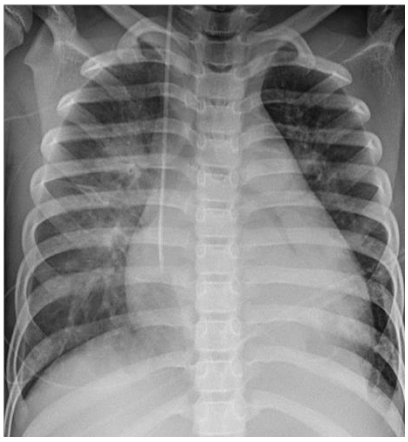
1–Chest x ray: Cardiomegaly, Kerley B line, plum oedema, pleural effusion.
2–Echocardiogram: Diagnostic
3–ECG to detect any arrhythmia

Other Laboratory: ABG– urea–electrolytes.

Chest x ray findings in heart failure :-







Treatment of HF: call for help & ABC (if needed), inform PICU

1. Put the infant in a semi sitting position.
2. O2 by mask – connect Pt to pulse oximetry.
3. Drugs decrease preload (Diuretics as furosemide) – decrease afterload (V. D as hydralazine) or both (ACEIs as captopril)
4. Drugs increase myocardial contractility (Digitalis)، or positive inotropes in cardiogenic shock
5. Care for feeding – NGT Feeding (Tachypnoea), High calorie diet.
6. Rx underlying cause.

Preload reducing drugs: غير مطلوب حفظ الجرعات

	mechanism of action	dose	common S/E
Furosemide (Lasix)	Loop diuretic (NaK2CI)	1mg/kg/dose (4–6 hr) intravenous	hypo K, hypo Na, Hypovolemia, M Alkalosis
Hydrochlorothiazide	DCT (NaCl)	1–2 mg/kg/day (8–12 hr) Oral	Hypo K, M alkalosis
spironolactone	Collecting ducts (aldosterone)	1–2 mg/kg/day (8–12 hr) Oral	Hyperkalaemia Gynecomastia

afterload reducing drugs:

	mechanism of action	dose	common S/E
Captopril	Short–acting ACE	0.1–6 mg /Ks/clay (8 hr) oral	Cough, rash, neutropenia, decreased GFR, hypo K.
Hydralazine	vasodilator	0.2–0.4 mg/kg/dose IV 1 mg/kg/6hrOral	Tachycardia Nausea, vomiting drug–induced SLE
Enalapril	Short–acting ACE	Single dose	Cough, rash, neutropenia, decreased GFR, hypo K.

Digitalis:

	action/dose	indication & CI	S/E
digoxin	↑Contractility + Conductivity Diuretic effect Dose: 0.005–0.01 mg/kg/d in 2 divided doses	Indications: • HF (with impaired contractility, • Atrial arrhythmias Contraindications: any Obstructive lesions, e.g, HOCM	digitalis toxicity

DIGITALIS TOXICITY: mcq

Manifestations	Factors increases risk of digitalis toxicity	treatment digitalis toxicity	Indications of serum digoxin level
1. GIT: Anorexia, nausea, vomiting, diarrhoea 2. CVS: Arrhythmias 3. CNS: Headache, visual disturbance (coloured vision) 4. Allergy, gynecomastia	1–HypoK &hyperCa 2–Diuretics(furosemide ,thiazides),verapamil, B–blockers 3–Renal impairment 4–Rheumatic & viral myocarditis 5–Preterm infant	1–STOP digitalis 2–Correction of hypoK& hyper Ca 3–TTT of arrhythmias: Phenytoin & Lidocaine 4–digibind{digitalis Ab}	1–Toxicity 2–Inadequate response 3–Impaired renal functions 4–Ingestion (accidentally) د عبدالرحمن الجعكي

Other drugs used in heart failure management:

1. Positive inotropes:
   - Indications: Cardiogenic shock
   - Drugs Used & administration: Cardiogenic shock:
2. Phosphodiesterase inhibitors, e.g, amrinone, milrinone
3. Beta blockers

Complications of HF:

1. Cardiac Arrhythmia & Thrombo–embolism
2. Repeated chest infections & Pulmonary oedema
3. Liver dysfunction & Renal failure
4. Electrolyte Imbalance
5. Failure to thrive – low weight and height

CARDIOMYOPATHY

	Dilated (most common)	Hypertrophic	Restrictive (very rare)
Causes	1. Idiopathic (>60%) 2. Past Viral 3. Familial (AD–AR–XL) 4. Metabolic (organic acidemia, FFA, oxidation disorder) 5. 2nd to systemic disease 6. Nutritional disease 7. DMD	1. Idiopathic 2. Familial (AD) 10–25% • Chest Pain • Sudden death 3. Obstructive CHD(AS) 4. IDM 5. Metabolic • GSD II (pomp's) • MPS • fabay disease	1. Idiopathic 2. Sarcoidosis 3. Amyloidosis 4. MPS 5. Malignancy
pathophysiology Dysfunction	Decrease contractility (systolic dysfunction)	• Decrease compliance (decrease filling) • Diastolic dysfunction	• Decrease Compliance (decrease filling) • Diastolic dysfunction
C/P	• Gradual onset S/S HF • Arrhythmia	• Asymptomatic • Decrease COP • Dyspnoea –Chest pain	• Decrease COP (easy fatigue, dizziness/syncope)
investigation	• CX Ray: Large Heart • Echo: Dilated Vt thin Wall	Echo: • Large heart • Thick Wall • Decrease Ejection fraction	Echo: Small or Normal Size
Rx	1. Rx of HF, arrhythmias 2. Heart Transplantation (most common cause)	1. Avoid severe physical activity 2. B–blockers (***Digitalis are CI***) 3. Antiarrhythmic drugs in case of arrhythmia. 4.Cardiac transplplantation	1. Rx of HE: Diuretics, ACE inhibitors, Digitalis 2. Anticoagulants to prevent thrombus formation in the dilated atria 3. Antiarrhythmic –Cardiac transplantation

MYOCARDITIS

Definition: Is inflammation of the myocardium leads to necrosis & fibrosis → weakness of the heart → impaired contractility → HF

- Myocarditis may be associated with pericarditis or endocarditis
- Myocarditis cane be asymptomatic, acute –fulminant –chronic
- Affect all age group including neonates

Causes:

1. Infectious
   - **Viral:** ***Coxsackievirus*** A&B, Adenovirus, EBV, MMR, VZ, HIV.
   - Bacteria: Diphtheria –TB– Rickettsia
   - Fungal: Histoplasmosis,
   - Parasitic: Trypanosoma –toxoplasmosis.
2. Toxin: scorpion venom
3. Collagen Vascular Disease: SLE, ***Kawasaki disease***, JIA
4. Idiopathic.
5. metabolic: GSD, MPS
6. Familial/Hereditary
7. Others
   - Drugs (**Adriamycin** –cyclophosphamide)
   - Diet (**Beriberi, Selenium** deficiency, KWO)
   - Endocrine (hypo or hyper T3T4)
   - Blood disease (thalassemia, leukaemia)

Post viral myocarditis:

Causes: Coxsackievirus B, Adenovirus, EBV, MMR, VZ, HIV

C/P:

- **H/O viral illness URTI** 10–14 days back
- S/S of acute HF (breathlessness, sweating, poor feeding, RD + 3T&1C)
- Arrhythmia, cardiogenic shock
- muffled heart sounds
- Mortality rate is high.

Acute myocarditis: Direct tissue damage (common in neonate)

Chronic myocarditis: Immune-mediated damage → Dilated cardiomyopathy (older child)

Diagnosis:

- CX Ray: Cardiomegaly
- **Echo: Diagnostic**
- CBC, ESR
- Cardiac Enzyme (CK, LDH) High
- **Viral Studies serology PCR**
- ECG: Arrhythmia

Treatment:

- Supportive + ICU care
- R_x of HF
- R_x of Arrhythmia.
- Intravenous immunoglobulin ???
- Steroid!!
- Heart Transplantation in refractory cases

RHEUMATIC FEVER

Definition:

- Is multi systemic autoimmune inflammatory disease following URTI with group A beta haemolytic strep
- Affecting (Heart – joints – skin – CNS)

Epidemiology:

- Age: 5–15 years
- Sex: more in ♀
- High % in developing countries with low socioeconomic status.

Pathogenesis

Autoimmune following ***upper respiratory tract infection*** with group A B-haemolytic streptococci after a latent period of 2–3 weeks (several months in rheumatic chorea)

Mechanism of tissue injury:

- **Molecular mimicry** between strep & tissue antigens.
- **Cross-reactivity** between strep & tissue antigens.

Pathology: (Rheumatic fever licks the joints & bites the heart)

Exudative: In Joints, Pericardium & Basal ganglion, heal without fibrosis

Proliferative: (Aschoff nodules): In endo–myocardium & skin, heal with fibrosis

Diagnosis: Modified jones criteria

Major Criteria	Minor Criteria	Evidence of Recent strep infection
1. Poly Arthritis 2. Carditis 3. Chorea 4. Erythema marginatum 5. SC nodules	1. Fever ≥38.5°c 2. Arthralgia 3. Prolonged PR 4. High WBC, CRP, ESR	1. +ve throat C&S 2. ASO titer increase 3. Recent scarlet fever 4. Anti–DNase B 5. Antistreptokinase Ab

Interpretations of Jones criteria: Diagnosis depends on:

Two major OR one major +2 minor criteria

+ evidence of recent streptococcal infection

- Diagnosis based on two major criteria is stronger than that based on 1 major & 2 minor criteria
- Exceptions: Chorea, Late–onset carditis, rheumatic recurrence in patients with documented RHD
- Fever>39.5° C is very unusual in rheumatic fever Arthralgia Should Not be considered a minor criterion in patients with arthritis
- Leukocytosis, ESR & CRP are all considered one minor criterion

1. Polyarthritis: 75% Common (50% illustrated textbook 6th edition) more in adult

- Affecting Large joints (knee– Ankle – wrist– Elbow)
- Usually earliest symptoms.
- Polyarthritis
- Migratory from Joint to other (Flitting)<1 week in a joint, leaving the joint completely free.
- The affected joint is red, hot, tender, swollen with limitation of movements.
- Non–destructive.
- Dramatic response to salicylates.
- Spontaneous resolution (even without ttt)

2. Carditis: 50%, (60 % illustrated textbook 6th edition)

- Is the most serious
- May Late onset
- Pancarditis: common in children
    1. Pericarditis
        - Chest pain increase respiration
        - Pericardial Rub
        - Pericardial Effusion
    2. Myocarditis
        - Muffled heart sounds
        - HF
        - Increase HR
    3. Endocarditis
        - Lt side > Right side
        - Mitral > aortic (acute)

- **ACUTE stage:**
  - Mitral regurge: Harsh pansystolic murmur propagating to axilla (damage of the cusps)
  - Mitral Stenosis: Carey–Coombs murmur: Mid–diastolic due to MS (edema of the cusps)
- **Chronic stage** (fibrosis): mitral stenosis

3. Sydentrams chorea: 10% (20 % illustrated textbook 6th edition)

- More in ♀
- affection of basal ganglia
- May be the only presentation of RF
- Usually late onset (months) >2–6 m

C/P

- Chorea decrease sleep – increase emotional stress
- Hypotonia
- Emotional Lability
- Resolve spontaneously, without residuals

4. Erythema Marginatum: (<5%)

- Large erythematous macules with pale canter "maple like outline"
- Non–Pruritic
- Site trunk – proximal parts of the limbs
- Uncommon, early manifestation
- Recurrent – Evanescent
- Increase "Hot "– decrease "Cold"

5. Subcutaneous Nodules: <1% (<10% % illustrated textbook 6th edition)

- Over bony prominences (Elbow– knee) on extensor surface.
- Hard – non tender – round nodules
- Usually associated with ***severe carditis***

Investigations:

1. Acute phase reactants
   - CBC
   - CRP
   - ESR
2. Evidence of Strep Infection
   - ASO Titer >300 Units
   - Anti–Streptokinase Ab
   - Anti DNAse B Ab
   - Throat Culture

3. Cardiac Assessment
    - CX Ray
    - ECG
    - Echocardiogram

Treatment:

- Supportive
    - Rest: Only in Pt with severe carditis or arthritis
    - Diet: Fluid restriction in case with HF
- Specific
    - **Arthritis or mild carditis: salicylates** (100 mg/kg/day 2 weeks then 75 mg/kg/day for 4–6 weeks) [naproxen!!]
    - **Severe carditis** with Cardiomegaly ***Steroid*** prednisolone (2 mg/ kg)
    - Disease activity can be monitored with serial ***CRP*** measurements.
    - **Chorea**: psychological, educational and social support until it resolves. Or {phenobarbitone – haloperidol}
    - Penicillin to **Eradicate** GAS carriage.
    - R_x of **complication** (HF– Arrhythmia)

Prevention:

Rheumatic fever now is rare in developed word why? B/C Improvement of sanitation – decrease overcrowding living condition – R_x of strep infection

- Early diagnosis of strep infection, proper treatment strep infection with Oral penicillin V for 10 days course.
- Prevention of recurrence of Rh fever
    - IM benzathine penicillin / 3–4 weeks (most effective)
    - Duration: either 10 years after the last episode of acute rheumatic fever or until the age of 21 years (If server carditis: lifelong prophylaxis)

Complication:

1. Early: HF, arrhythmias
2. Late: RHD, rheumatic activity (recurrence), infective endocarditis.

INFECTIVE ENDOCARDITIS

Definition: Infection of the endocardial surface of the heart (valvular or mural)

(All children of any age with congenital heart disease (except secundum ASD), including neonates, are at risk of infective endocarditis).

Pathogenesis:

1. **Structural Heart Disease**
   - Congenital (All except ASD secundum) as VSD– COA, PDA– TOF
   - Acquired:
     - ✓ Prosthetic Valve.
     - ✓ Previous IE.
     - ✓ RHD.
     - ✓ Mitral valve prolapse with MR.
2. **Bacteraemia**
   1. α–haemolytic streptococcus = (Streptococcus viridans) (The most common causative organism) 50%
   2. Staph. aureus, epidermis (40%)
   3. Enterococci (GIT–GU) HACEK (Haemophilus, Actinobacillus, Cardio bacterium, Eikenella, Kingella).
   4. Pseudomonas (IV drug use)
   5. Fungal (immune deficiency)

Pathophysiology of Subacute endocarditis

The congenital or acquired abnormal valve cause flow disturbances that injure the endocardial lining of the heart valves or wall

Dental work or manipulation of either the gastrointestinal or genitourinary tracts surgery can cause bacteremia

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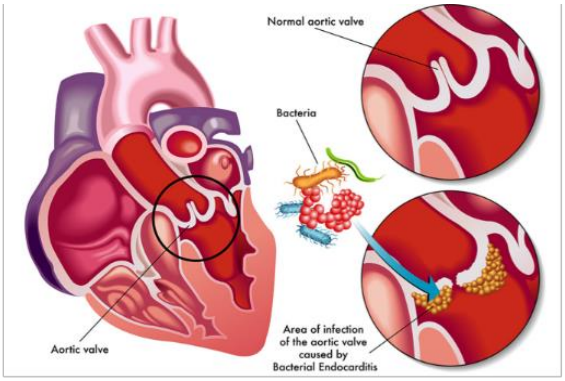
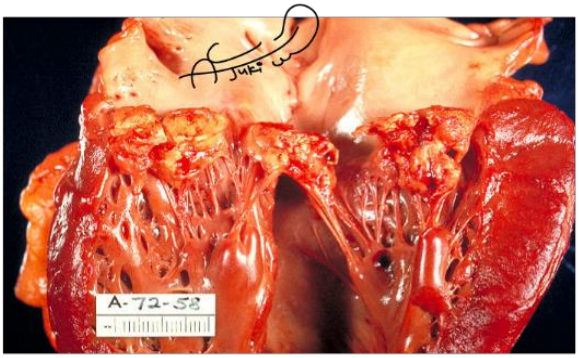
The pathogen seeding to injured endocardium &+++ inflammation

Vegetation grow, erode the valve, and may create myocardial abscesses

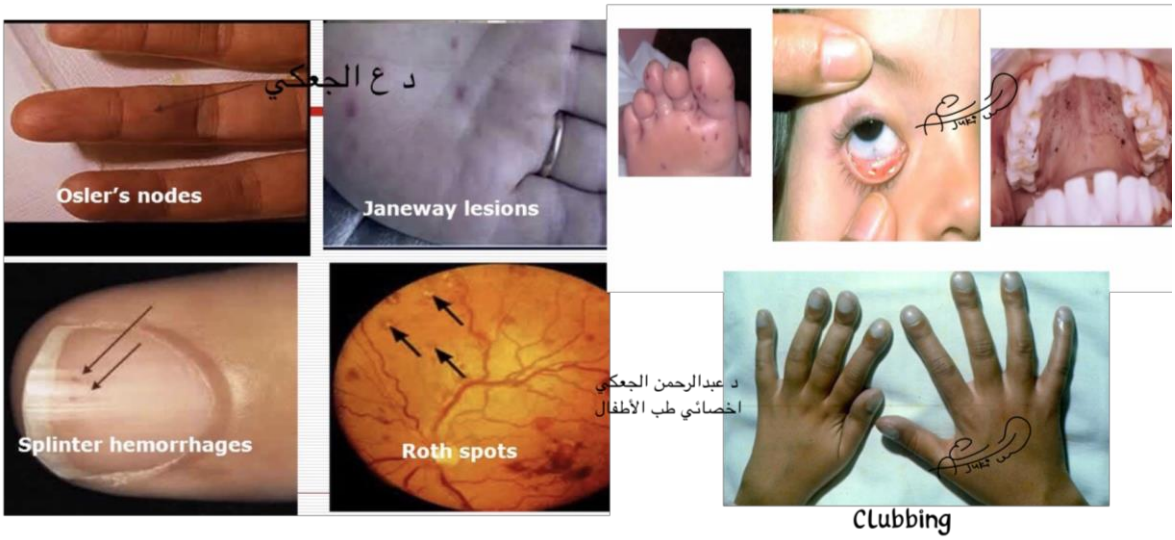
Fragments can break off as emboli causing embolic manifestations &immune complex manifestation occurs later .





Clinical Picture: It may be difficult to diagnose, high suspension rate is important

- General features
  1. Fever (high grade), Chills
  2. Anorexia, fatigue, wt loss, Pallor, Arthralgia (Arthritis)
  3. Pulse: high HR
  4. Hands:
     - Clubbing (**late**)
     - splinter Haemorrhage under nails.
     - Osler's nodes n (painful).
     - Janeway nodules (Painless)
  5. Eyes (subconjunctival Hge– Roth spot in the retina).
  6. Splenomegaly (70%)
  7. Glomerulonephritis GN (**microscopic hematuria**)
  8. Hemiplegia, Hge
  9. Necrotic Skin Lesions



Signs of Infective endocarditis

- Cardiac features
  1. New murmur
  2. Change of the character of the already present murmur.
  3. S/S HF (due to valve damage)

Diagnosis:

1. CBC (high WBC) (plt ,Hb)
2. CRP– ESR– high (useful to monitor response to treatment)
3. Blood C&S (Multiple before antibiotics at least 3 samples)
4. Echocardiography: vegetation (absence of vegetations does Not exclude IE)– valve lesion – Abscess
5. ECG
6. Urine for microscopic haematuria
7. RFT, Electrolyte, low C_3 – C_4

- Duke Criteria:

Major Criteria	Minor Criteria
1. ≥ 2 +ve BL C&S 2. +ve Echo Finding • Vegetation • New Valvular damage	1. Fever 2. Embolic manifestation 3. Immune Complex Disease 4. Predisposing factors 5. Single +ve blood C&S

Interpretations Of Duke Criteria:
•Definitive IE: (2 major) OR (1. major +3 minor) OR (5 minor)

Treatment:

1. **Empirical antibiotic IV** for **6 weeks**
   High dose Penicillin + Gentamicin IV or
   Penicillin + Ceftriaxone IV
2. Surgery in case of abscess or large vegetation or Infected prosthetic material

Prophylaxis:

1. **Good dental – oral hygiene** (The **most important** factor in prophylaxis)
2. **Antibiotic prophylaxis!!** in UK (for high & moderate risk category before surgery as Dental procedures, e.g, tooth extraction & adenotonsillectomy

Amoxicillin single dose 30–60 minutes before the procedure, or injection ampicillin or (ceftriaxone single dose –clindamycin –vancomycin used also)

Risk category for IE:

High–risk category	Moderate–risk category	Negligible–risk category
1. Complex cyanotic CHD as TOF, TGA 2. Repaired CHD with residual defects 3. Prosthetic valves 4. Previous IE	1. Most other CHD, e.g, VSD, PDA, ASD primum, CoA 2. RHD 3. MVP with MR	1. Secundum ASD 2. Repaired VSD & PDA > 6m 3. Pacemakers &ICD 4. MVP without MR 5. Coronary artery bypass

CARDIAC ARRHYTHMIAS

Sinus arrhythmia is normal in children & is detectable as change in heart rate with respiration (Increase insp – decrease Exp) up to 30 b/min

Supraventricular Tachycardia (SVT):

- Is the most common childhood symptomatic arrhythmia, HR is rapid (250–300 b/min)
- Its due to re–entry 2 pathways are involved
  - ➢ One of them is AVN
  - ➢ The other is accessory pathway

Aetiology:

1. **Idiopathic**" MC "
2. Myocarditis – cardiomyopathy
3. CHD as Ebstein anomaly, single ventricle. (**structural heart defects are rare**)
4. Drugs, Electrolyte imbalance, post cardiac surgery.

Clinical Picture: Attacks or Episodes – Few Seconds

1. Intrauterine SVT: Hydrops fetalis
2. Neonate & Infants: Sudden pallor – irritability – vomiting, **Heart failure** if prolonged.
3. Children: Palpitation – chest pain – pallor – Syncope

Diagnosis:

1. ECG (very **Rapid HR – No p wave– narrow QRS**)
2. 24 hours ECG (Holter ECG)
3. Echocardiography (to exclude structural heart defect & should be performed both to review cardiac anatomy as well as evidence of poor function)

Treatment: call for help –ABC IF unstable, connect pt to 12 leads ECG

Hemodynamically:

1. Stable:
   - ✓ **Vagal stimulation "80% successful"**
     - Infants: (ice – water on face)
     - Child (Valsalva manoeuvre) or (carotid massage!!)
   - ✓ **Drugs: Adenosine** (DOC)0,1 mg /kg, safe and effective, it blocks AVN, IV Bolus followed by flush with normal saline
     - Can be repeated if not effective (0,2 mg /kg)

2. **Unstable**
    - Cardioversion DC Shock [0.5– 2 J/kg]
    - Can be repeated 3 times
    - If not effective [amiodarone Iv infusion]

After discharge: send to cardiology OPD for echo & follow up + antiarrhythmic drugs

Prevention:

- Maintenance therapy as Sotalol – Propranolol – Flecainide to prevent recurrence
- Treatment usually stops at 1 year of age in SVT (no further attacks after infancy)

- In SVT: ECG Normal between attacks
- If there is Short PR interval + **Delta** wave: **Wolf – Parkinson – White–syndrome (WPW)**
- Ablation of accessory bundle in **WPW** (pre–excitation) syndrome

Congenital complete heart block (rare)

- Maternal SLE: anti–Ro or anti–La antibodies → cross placenta → prevent normal development of the conduction system in the developing heart, fibrosis of the AV node → **complete heart block**
- Either leads to fetal hydrops, death in utero and heart failure in the neonatal period. However, most remain symptom free for many years, but a few become symptomatic with presyncope or syncope

Treatment: permanent pacemaker.

Long QT syndrome: (LQTS) . Very rare اقل اهمية

Etiology:

A) **Congenital** (defective ion channels):

- Romano– Ward syndrome (AD)
- Lange–Nielsen syndrome (AR) → **LQT** + deafness

B) **Acquired**:

- Drugs: Erythromycin –Ketoconazole.
- Electrolyte Disturbance: hypo K, hypo Ca, hypo Mg.

C/P:

- Syncope (often precipitated by exercise) & palpitation
- Seizures (misdiagnosed as epilepsy)
- Sudden death (family history of sudden unexplained death)

ECG: Prolonged QT intervals in ECG (> 0.47 sec)

Treatment: Propranolol , ICD

SYNCOPE

- **It's a transient loss of consciousness.**
- It's caused by transient impairment of brain O_2 delivery –>decrease cerebral perfusion
- It is common in **adolescents** but usually **benign – Rarely due to heart disease**

Causes:

1. **Neurologically mediated** (vasovagal syncope) most common
   - Characterized by prodrome (nausea, sweating, headache) followed by LOC
   - Triggers: Prolonged Standing, sight of blood, fright, crowded places
2. **Cardiac syncope**
   - Due to arrhythmia SVT– HB – AS – HOCM, LQTS
   - Symptoms
     Symptoms on exercise
     Family history of sudden unexpected death
     Palpitation

Management:

- Good history & Check (BP– murmur– pulse– sign of Marfan syndrome)
- 12 lead ECG
- R_x Underlying cause

Pulmonary hypertension: is defined as a mean pulmonary artery pressure (PAP) **greater than 25 mm Hg at rest.** (mean PAP of 8 to 20 mm Hg at rest is considered normal)

Causes:

Pulmonary arterial hypertension	Pulmonary venous hypertension	Pulmonary hypertension with respiratory disease
1. Idiopathic: sporadic or familial 2. Post–tricuspid shunts (e.g. VSD, AVSD, PDA) 3. Persistent pulmonary hypertension of the newborn	1. Left–sided heart disease 2. Pulmonary vein stenosis or compression	1. Chronic obstructive lung disease or 2. bronchopulmonary dysplasia in preterm infants 3. Interstitial lung disease 4. Obstructive sleep apnoea

Treatment:

1. Medical: Many medical therapies are now available, (e.g. inhaled **nitric oxide,** intravenous magnesium sulphate and **sildenafil**) or (intravenous prostacyclin or inhaled iloprost), anticoagulant

2. surgical: **heart lung transplantation** in advanced cases

Summary

Left Heart outflow obstruction in sick infant (Duct dependent):

1. **Critical COA:** Circulatory collapse, Absent femoral pulse – ABC, PG infusion
2. **Interruption of aortic arch:** Circulatory collapse, Absent femoral & Lt brachial pulse– ABC PG infusion
3. **HLHS:** Circulatory collapse, all peripheral pulses absent – ABC PG infusion

DD of Oligemic lung	DD Of plethoric lung
1. Pulmonary Atresia 2. Tricuspid Atresia 3. Pulmonary Stenosis or HTN 4. TOF 5. Esinmenger syndrome 6. Ebstein Anomalies	1. VSD 2. PDA 3. ASD 4. AVC 5. TAPVR 6. TGA

Chest X Ray appearance	CHD
1. Boot Shaped heart → 2. Egg on side heart → 3. Rib notching, 3 sign→ 4. Snowman appearance→ 5. Box shaped heart→	• TOF • TGA • COA • TAPVR • Ebstein Anomalies

Definition "Radiology"

Situs Solitarius →	• Normal position of organs
Laevocardia→ Situs Inversus→	• Heart in normal position (Apex on Lt side) • All organs in the opposite site (heart apex stomach: Rt) (Liver: Lt)
Dextrocardia →	• Only Heart in the opposite (Apex: Rt)

Duct Dependent CHD:

1. TA-PA
2. Ebestion Anomalies
3. TGA
4. HLHS
5. Critical COA – AS
6. Interrupted aortic arch

تنبيه:

هذا العمل تم بعد جهد كبير من قبل الدكتور الاخصائي: عبدالرحمن الجعكي، ويعتبر ثمرة سنين من خبرة ممارسة طب الأطفال، كل ذلك كان ممزوج بحنكة العطاء الاكاديمي للمادة.

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